

Phase 6 — Operations & Engagement Spec

PREPARED BY Energy Hippo implementation team

The operational layer plus scheduled report delivery to **100+ users**. Closes the remaining reports: Project (4), Alarm, Audit, ActivityTracker, TaskLog, GatewayStatus, EnergyAllInsights, HMRConfiguration, bill_importer/my_imports — and adds the MFR delivery layer over all 92 deliverable reports.

Entry path (see [entry-path-and-gaps.pdf](#)): project tracking, alarm **thresholds** (WPF), task **schedules** (web), workflow authoring (WPF ConfigurationTool), MFR save/dashboard/delivery, and handheld config are **UI-DIRECT**. Audit/activity/status/notes are **SYSTEM-GENERATED** (free by using the app). Alarm **events**, **task-log** history, and AP/regression results are **UI-THEN-RUN** (the Tasks EngineWorker / jobs must be running). **Gaps**: **EnergyAI** insights are **EXTERNAL**-only (B2); **GreenButton** import has no UI (A8); MFR **external-email** recipients are a missed screen — reach **100+ via groups** (A7). Project types/categories/activities are reference data (no create UI).

Source-model extension

Add: project (type, category, status, capital/operating/incentive cost), project_activity (type, dates, hierarchy), project_event (type, milestone); alarm_point (point, alarm_type, priority, thresholds), alarm_event (point, ts, value, priority, ack, resolved); audit_event (user, action, table, ts); task_schedule (task, kind, cron/interval, next_run), task_log_entry, workflow (nodes); mfr_saved_report, mfr_dashboard, mfr_delivery (schedule, formats), mfr_delivery_recipient (kind, user/group/email); energyai_insight; hmr_meter/hmr_read_type/hmr_reading; import_source/ import_mapping.

A. Project tracking

A campus sustainability portfolio (Project → Activity → Event, verified tables exist):

project	category	capital	incentive	M&V
Library LED Retrofit	Lighting	\$120k	\$25k rebate	baseline b1_nlu_* → post lighting kWh
Central Plant Chiller Upgrade	HVAC	\$640k	\$80k	plant electric kWh per chilled-water ton-hour produced
Cedar Row Solar Expansion	Renewable	\$410k	ITC	solar generation vs prior
Admin Hall Recommissioning	Controls	\$55k	\$10k	gas + electric baseline

Each with activities (Design → Procurement → Install → Commissioning → **M&V**) and financial events (cost, savings) tied to hierarchy nodes. Lights up ProjectDetail/Summary/Finance + rp_ApplyRateSchedule.

B. Alarms

Alarm-enabled points with `PriorityLevels` (1 Informational ... 5 Emergency):

point	alarm	priority	raised
Student Center demand kW	high-demand threshold	4 Critical	summer event spikes
Chiller enable (digital)	chiller-fault	3 Important	occasional
Science Center interval	data-gap	2 Warning	the <code>missing_interval_backfill</code> window
Cedar Row water kgal	high-usage	2 Warning	occupancy spikes

Each with `raised` → `acknowledged` → `resolved` `alarm_event` rows across the span → `rp_AlarmDataMap`.

C. Audit & activity logs

Enable audit; generate a realistic trail across the 19 users: bill entry/approval, rate edits, hierarchy changes, report runs, logins — including the **disabled user (casey.holt) failed-auth** attempt and the **cross-company auditor (quinn.roberts)** views. Lights up `rp_Audit` + `rp_ActivityTracker`.

D. Tasks / workflows

`task_schedule` + `task_log_entry` (TaskType: 3 EEM Gateways, 6 Bill Importer, 4 Report, 5 MFR, 2 Standard Alerts, 1 System):

- Schedules for the 15 gateway profiles, the bill importer, and report/MFR delivery (interval + cron).
- TaskLog run history across 2024–2025 (successes + a few failures) → `rp_TaskLog`, `rp_GatewayStatus`.
- Two **workflows**: (1) download → bill-import → validate → notify finance; (2) gateway read → interval publish → weather regression → alert.

E. MFR saved reports & delivery (100+ users)

Scale default (owner decision still open): a **Monthly Building Energy Report** delivered to a **building-representative roster of ~120 users**, reached via **group** recipients (most efficient). Roster defined by a deterministic generation rule (120 users across ~12 building-rep groups, ~10 each), not enumerated.

- `mfr_saved_report` library: Monthly Building Energy, Weekly Cost Rollup, Executive Sustainability Summary, Daily Operations — each with saved parameters.
- `mfr_dashboard`: Facility Operations, Executive Overview.
- `mfr_delivery` schedules: Monthly (cron `0 7 1 * *`) → the 120-user building-rep groups (PDF); Weekly (cron) → finance group + 2 external emails (PDF+Excel); Daily (interval) → ops individuals. Recipients via `mfr_delivery_recipient` (kind group/user/email). Executed by the MFR Reports task.

F. EnergyAI

Pre-generated `energyai_insight` rows for two well-instrumented load points (Science Center electric, Student Center electric): anomaly detections (e.g. weekend baseload spike), forecast vs actual, and savings opportunities, with energy/demand/cost metrics → `rp_EnergyAIInsights`.

G. HMR (handheld meter reading)

Explicit `hmr_meter` route/cycle/sequence + `hmr_read_type` (dials/decimals/ direction/demand) tied to the existing **Neptune (Cedar Row B water master, and the Cedar Row apartment water submeter route)** and **MVRS (Student Center gas)** handheld uploads, with `hmr_reading` rows → `rp_HMRConfiguration`, `hmr_export`.

H. Importers / file intake

- **Bill Importer:** an `import_source` + `import_mapping` (provider/account/ commodity) + sample **BIF/CSV/Excel** bill files, including the shipped **bad-data validation sample** and the **2026-01 fiscal-gap** rejection file (Phase 0 fixture) → `bill_importer`, `my_imports`.
- **GreenButton:** one ESPI XML interval feed for a metered building.
- **MyUpload/MyImport:** a JSON upload handler + a `myimport_intervaldata` / `myimport_MeterReadings` template run.

expectedAssertions

- `projects_with_finance` — Project Detail/Summary/Finance render with capital/ incentive/savings; M&V activities link to baselines.
- `alarms_lifecycle` — `rp_AlarmDataMap` shows raised → ack → resolved across priorities; the data-gap alarm aligns with the backfill window.
- `audit_trail` — `rp_Audit`/`rp_ActivityTracker` show edits + the disabled-user failed auth + cross-company auditor views.
- `task_history` — `rp_TaskLog`/`rp_GatewayStatus` show scheduled runs incl. failures; workflows complete.
- `mfr_delivery_100plus` — the Monthly delivery fans out to ≥100 recipients (verifiable via the delivery task log) in PDF.
- `energyai_insights_present` — `rp_EnergyAIInsights` returns insights with metrics.
- `hmr_config_renders` — `rp_HMRConfiguration` shows the Neptune/MVRS routes with read types.
- `importers_ingest` — sample BIF/CSV/Excel + GreenButton + MyUpload all ingest; the bad-data and 2026-01 files are rejected with the expected messages.
- `every_report_nonempty` — with all phases loaded, each of the 115 reports has at least one scenario that renders it non-empty.